Visvesvaraya Technological University

Belgaum, Karnataka- 590014

Third Year

A Mini-Project Report

On

“Your Project Title Goes Here”

Submitted in the partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)

Submitted by

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Under the Guidance of

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Associate Professor, Dept. of CSE (Data Science), DSCE

2025–2026

DEPARTMENT OF CSE (DATA SCIENCE)
DAYANANDA SAGAR COLLEGE OF ENGINEERING
SHAVIGE MALLESHWARA HILLS, KUMARASWAMY LAYOUT, BANGALORE-78

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout
Bangalore-560078

Department of CSE (DATA SCIENCE)

2025–2026

Certificate

This is to certify that the Mini Project Work entitled **"Your Project Title Goes Here"** is a bonafide work carried out by **Student One Full Name (1DSXXCDXXX), Student Two Full Name (1DSXXCDXXX), Student Three Full Name (1DSXXCDXXX) and Student Four Full Name (1DSXXCDXXX)**, in partial fulfilment for the VI semester of Bachelor of Engineering in CSE (Data Science) of the Visvesvaraya Technological University, Belgaum during the year 2025–2026. The Project report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

Signature of Project Coordinator	Signature of Guide	Signature of HOD
[Dr. Coordinator Name]	[Dr. Guide Name]	[Dr. HOD Full Name]

Name of the Examiners	**Signature with Date**
1.	
2.	

ACKNOWLEDGEMENT

First and foremost, we thank the Almighty for the blessings and guidance throughout this project.

We express our sincere gratitude to our Principal, Dr. Principal Name, for providing all necessary facilities.

We thank our Head of Department, Dr. HOD Full Name, for the constant motivation and support extended to us.

We extend our heartfelt thanks to our guide, Dr. Guide Name, for the valuable guidance and suggestions throughout the project.

We are grateful to all teaching and non-teaching staff of the Department of CSE (Data Science) for their direct and indirect support. We also thank our family and friends for their encouragement throughout.

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Chapter 1

Introduction

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Chapter 2

Literature Survey

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Chapter 3

Research Gap Identified

From the literature survey, the following research gaps are identified:

- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
- Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat duis aute irure dolor.
- Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum sed ut perspiciatis.
- Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione.
- At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores.
- Therefore, there is a need to develop a system that addresses the above limitations and demonstrates practical applicability in a real-world context.

Chapter 4

Architectures

4.1 High Level Architecture

The high level architecture of the proposed system is shown in Figure 4.1.

Figure 4.1: High Level Architecture of the Proposed System

4.2 Use Case Diagram

The use case diagram shown in Figure 4.2 represents the interaction between the user and the proposed system.

Figure 4.2: Use Case Diagram of the Proposed System

4.3 Data Flow Diagram

4.3.1 Level 0 DFD

Figure 4.3: Level 0 Data Flow Diagram

4.3.2 Level 1 DFD

Figure 4.4: Level 1 Data Flow Diagram

4.4 Detailed System Architecture

4.4.1 Input Processing Layer

Figure 4.5: Detailed Architecture – Input Processing Layer

4.4.2 Model and Core Processing Layer

Figure 4.6: Detailed Architecture – Model and Core Processing Layer

4.4.3 Output and Recommendation Layer

Figure 4.7: Detailed Architecture – Output and Recommendation Layer

Chapter 5

Project Objectives

- To develop an intelligent system for automatic detection and classification using domain-specific data.
- To design and implement a scalable preprocessing and feature engineering pipeline.
- To build and evaluate a deep learning model achieving competitive performance on the selected dataset.
- To incorporate Explainable AI methods for transparent and interpretable predictions.
- To develop a recommendation module generating actionable insights using a retrieval-augmented generation approach.
- To create an interactive dashboard for visualizing predictions, explanations, and recommendations.
- To evaluate the system against baseline approaches using standard performance metrics.
- To build a modular and extensible framework suitable for future real-world deployment.

Chapter 6

Timeline using Gantt Chart

6.1 Timeline using Gantt Chart

Figure 6.1 illustrates the project timeline across the academic semester. The phases include literature review, dataset preprocessing, model development, explainability integration, recommendation engine development, and dashboard implementation.

Figure 6.1: Project Gantt Chart

Chapter 7

Hardware and Software Requirements

7.1 Hardware Requirements

- Personal Computer or Laptop (minimum 8 GB RAM)
- Intel i5 processor or higher
- Minimum 256 GB storage
- Internet connectivity

7.2 Software Requirements

- Operating System: Windows / macOS / Linux
- Programming Language: Python 3.10 or higher
- Development Environment: Visual Studio Code / Jupyter Notebook
- Deep Learning Frameworks: TensorFlow or PyTorch
- Libraries: NumPy, Pandas, Scikit-learn, Matplotlib
- Explainable AI Tools: SHAP, LIME
- API Framework: FastAPI or Flask
- Vector Database: ChromaDB
- Version Control: Git and GitHub

Chapter 8

Sample Code

8.1 Sample Module Implementation

The following snippet demonstrates a core component of the proposed system.

Listing 8.1: Sample Preprocessing Utility

```
import numpy as np
import pandas as pd


def load_and_preprocess(filepath: str, target_col: str):
    """Load CSV and return normalized features and labels."""
    df = pd.read_csv(filepath).dropna()

    X = df.drop(columns=[target_col])
    y = df[target_col]

    numeric_cols = X.select_dtypes(include=[np.number]).columns
    X[numeric_cols] = (
        X[numeric_cols] - X[numeric_cols].mean()
    ) / X[numeric_cols].std()

    return X, y


if __name__ == "__main__":
    X, y = load_and_preprocess("data/sample.csv", target_col="label")
    print("Features:", X.shape)
    print("Labels:\n", y.value_counts())
```

Chapter 9

Project Demo Screenshots

Figure 9.1 shows the main dashboard of the proposed system.

Figure 9.1: Dashboard Interface of the Proposed System

Figure 9.2 shows the feature importance visualization from the explainability module.

Figure 9.2: Feature Importance Visualization

Figure 9.3 shows the recommendations generated by the system.

Figure 9.3: System-Generated Recommendations

Chapter 10

AI Tools Application

The proposed project utilized various AI-assisted development tools during research, design, implementation, and documentation.

10.1 OpenAI – ChatGPT

- Code generation for data preprocessing and model training
- Explaining deep learning and Explainable AI concepts
- Debugging machine learning pipelines
- Documentation and report writing support

10.2 Google – Gemini

- Research assistance and workflow planning
- Exploring multimodal learning and transformer architectures
- Architectural design and system workflow refinement

10.3 GitHub – GitHub Copilot

- Auto-completing repetitive code structures
- Writing model training and evaluation pipelines
- Debugging and improving code readability

Chapter 11

SDGs Met by Project with Proper Justification

- **SDG 3: Good Health and Well-Being** — The system supports health and well-being by enabling early detection and AI-driven decision support, providing personalized recommendations to improve user outcomes.
- **SDG 9: Industry, Innovation and Infrastructure** — The project applies advanced AI technologies including deep learning and Explainable AI to build an innovative decision-support framework for intelligent automation.
- **SDG 4: Quality Education** — The system and its documentation serve as a learning resource demonstrating practical integration of machine learning, explainability, and recommendation systems.
- **SDG 8: Decent Work and Economic Growth** — By automating classification and decision-making tasks, the system reduces manual effort and supports more efficient, sustainable workflows.

Chapter 12

Purpose of the Project

12.1 Purpose of the Proposed System

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. The proposed system aims to develop an interpretable end-to-end AI framework combining machine learning, Explainable AI, and retrieval-augmented generation.

12.2 Targeted Audience

- Domain experts who need to understand and validate AI predictions
- Researchers working on explainable AI and human-AI interaction
- Data scientists building production-grade ML pipelines
- Academic institutions using it as a reference implementation

12.3 Expected Outcome of the Project

- Accurate multi-class classification on domain-specific data
- SHAP-based feature importance explanations per prediction
- Contextual recommendations via retrieval-augmented generation
- Interactive web dashboard presenting all outputs

12.4 Research and Future Scope

- Integration with real-time data streams and edge devices

- Inclusion of additional data modalities
- Larger-scale real-world validation
- Mobile interface for end-user accessibility

Chapter 13

Research Paper and Plagiarism Report

13.1 Research Paper Details

As part of the project work, a research paper titled **"Your Full Research Paper Title Goes Here"** has been prepared, covering the proposed system architecture, methodology, experimental evaluation, and conclusion.

13.2 Plagiarism and AI Detection Report

The research paper was checked using Turnitin. The plagiarism and AI detection reports are attached in the Appendix and confirm a low similarity index satisfying originality requirements for academic submission.

Chapter 14

How to Use This Template

This chapter is a complete guide to using this template. No prior LaTeX knowledge is assumed.

14.1 Step 1 — Fill In Your Details

Open `main.tex` and find the section at the top marked **FILL THIS**. Update every line in that block. Nothing else in the file needs to be touched for the title page, certificate, header, and acknowledgement to work correctly.

- `\ProjectTitle` — Full title of your project
- `\HeaderTitle` — Short version used in the page header (keep it under 8 words)
- `\AcademicYear` — e.g. `2025--2026` (use two dashes for an en-dash)
- `\StudentOne` through `\StudentFour` — Full names of all team members
- `\USNOne` through `\USNFour` — Corresponding USNs
- `\GuideName` — Your guide's full name
- `\GuideDesignation` — Guide's designation and department
- `\ProjectCoordinator` — Project coordinator's name
- `\HODName` — Head of Department's full name
- `\PrincipalName` — Principal's full name

14.2 Step 2 — Compile Once to Check

Before touching anything else, compile to confirm the template builds on your machine. Run these four commands in your terminal from inside the project folder:

Listing 14.1: Full Compilation Sequence

```
pdflatex main.tex
bibtex main
pdflatex main.tex
pdflatex main.tex
```

You need to run `pdflatex` three times. The first pass builds the document and writes temporary files. `bibtex` processes the references. The second and third passes resolve cross-references and the table of contents. Skipping steps will cause **[?]** to appear instead of figure numbers or citations.

Using an editor instead: Overleaf, TeXstudio, and VS Code with the LaTeX Workshop extension all handle the multi-pass sequence automatically when you click Build. Overleaf is recommended if you have no LaTeX installation locally — just upload the entire folder and compile.

14.3 Step 3 — Folder Structure

Place your files according to the following layout:

- `images/VTULogo.png` — VTU logo (already included, do not rename)
- `images/DSCELogo.png` — College logo (already included, do not rename)
- `images/placeholder.png` — Used by all figures until you replace them. Copy any existing image and name it `placeholder.png` to keep compilation clean.
- `images/Diagrams/` — Put all architecture diagrams, DFDs, and use case diagrams here
- `images/screenshots/` — Put all dashboard and demo screenshots here
- `appendix/` — Put your plagiarism and AI detection PDFs here
- `references.bib` — All your references in BibTeX format

Important: File names are case-sensitive. `Dashboard.png` and `dashboard.png` are different files. Use lowercase names with underscores to avoid issues.

14.4 Step 4 — Adding Images

Every figure in this template currently points to `images/placeholder.png`. Once you have your actual diagrams and screenshots ready, replace the path inside each `\includegraphics` command with your real file path.

The full syntax for inserting an image is:

Listing 14.2: Image Block Syntax

```
\begin{figure}[H]
    \centering
    \includegraphics[width=0.75\textwidth]{images/Diagrams/your_image.png}
    \caption{Your caption text here}
    \label{fig:your_label}
\end{figure}
```

Controlling image size — change the `width` value:

- `width=\textwidth` — full page width (use for wide diagrams)
- `width=0.9\textwidth` — 90% of page width
- `width=0.65\textwidth` — 65% (good for portrait diagrams)
- `width=0.5\textwidth` — half width (good for small charts)
- `width=5cm` — fixed size in centimetres

Referring to a figure in text — use `\ref{fig:your_label}` and LaTeX will automatically insert the correct figure number:

Listing 14.3: Referring to a Figure

```
The architecture is shown in Figure~\ref{fig:your_label}.
```

14.5 Step 5 — Adding Lists

Bullet list (unordered):

Listing 14.4: Bullet List

```
\begin{itemize}
    \item First point
    \item Second point
    \item Third point
\end{itemize}
```

Numbered list (ordered):

Listing 14.5: Numbered List

```
\begin{enumerate}
    \item First step
    \item Second step
    \item Third step
\end{enumerate}
```

Lists can be nested by placing one `itemize` or `enumerate` block inside another `\item`.

14.6 Step 6 — Adding Code

Use the `lstlisting` environment. Set `language` to `Python`, `bash`, `TeX`, `SQL`, `Java`, `C`, or `C++`.

Listing 14.6: Code Block Syntax

```
\begin{lstlisting}[language=Python, caption={Your caption}, label={code:your_
    label}]
# your code here
print("Hello")
\ end{lstlisting}
```

Note: remove the space between \ and `end` in actual use — it is written that way here only to prevent the block from closing early.

14.7 Step 7 — Adding References

Open `references.bib`. Each reference is one block called a BibTeX entry. The template already contains sample entries for every common type. Copy the one that matches your source and fill in the fields.

Journal article:

Listing 14.7: Journal Article BibTeX Entry

```
@article{yourkey2024,
  author  = {Surname, Firstname and Second, Author},
  title   = {Title of the Paper},
  journal = {Name of Journal},
  volume  = {12},
  number  = {3},
  pages   = {100--115},
  year    = {2024},
  doi     = {10.xxxx/xxxxxx}
```

```
}
```

Conference paper:

Listing 14.8: Conference Paper BibTeX Entry

```
@inproceedings{yourkey2023,
  author    = {Surname, Firstname and Co, Author},
  title     = {Title of the Paper},
  booktitle = {Proceedings of the Conference Name},
  pages     = {45--52},
  year      = {2023},
  doi       = {10.1109/xxxxxx}
}
```

Website or dataset:

Listing 14.9: Website or Dataset BibTeX Entry

```
@misc{yourkey2022,
  author       = {Author, Name},
  title        = {Title of the Resource},
  year         = {2022},
  howpublished = {\url{https://example.com}},
  note         = {Accessed: January 2024}
}
```

The key (e.g. `yourkey2024`) is what you use to cite the reference in text:

Listing 14.10: Citing a Reference

```
According to Smith et al.~\cite{yourkey2024}, ...
```

The ~ before `\cite` adds a non-breaking space so the citation number never wraps onto a new line by itself.

14.8 Step 8 — Adding More PDFs to the Appendix

To attach an additional PDF (e.g. your actual research paper), add the following block inside the Appendix chapter at the bottom of `main.tex`:

Listing 14.11: Attaching a PDF to the Appendix

```
\section{Your Section Title}

Brief description of what this document is.
```

```

\includepdf[
    pages=-,
    pagecommand={\thispagestyle{mainstyle}}
]{appendix/your_file.pdf}
```

`pages=-` means include all pages. To include only specific pages, use `pages={1,3,5}` or a range like `pages={2-4}`.

14.9 Quick Reference Card

- **Bold text:** `\textbf{your text}`
- **Italic text:** `\textit{your text}`
- **Inline code / filenames:** `\texttt{your text}`
- **New section:** `\section{Section Title}`
- **New subsection:** `\subsection{Subsection Title}`
- **Line break in title page:** `\\`
- **En-dash (year range):** `2025--2026` renders as 2025–2026
- **Non-breaking space:** `~` (use before `\cite` and `\ref`)
- **Percentage sign:** `\%` (plain % starts a comment)
- **Ampersand:** `\&`
- **Comment (ignored by compiler):** start the line with `%%`

References and Bibliography

[1] First Author and Second Coauthor. Title of the journal paper goes here. *Name of the Journal*, 12(3):100–115, 2023.

[2] First Author, Second Coauthor, and Author Third. Title of the conference paper. In *Proceedings of the International Conference on Relevant Topic*, pages 45–52. IEEE, 2022.

[3] Firstname Surname and Author Another. Deep learning approach for classification of structured data. *Expert Systems with Applications*, 198:116–130, 2022.

[4] One Reviewer and Author Survey. A survey on explainable artificial intelligence methods for tabular data. *ACM Computing Surveys*, 55(7):1–35, 2023.

[5] Name Author and Author Co. Retrieval-augmented generation for knowledge-grounded decision support. In *Findings of the Association for Computational Linguistics*, pages 200–212, 2023.

Chapter A

Research Paper, Plagiarism and AI Detection Reports

A.1 Turnitin Similarity Report

The Turnitin similarity report begins from the next page.

Conference Paper Title*

*Note: Sub-titles are not captured in Xplore and should not be used

1st Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

2nd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

4th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

5th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

6th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

This document is a model and instructions for LaTeX. Please observe the conference page limits.

II. Ease of Use

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

III. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

Identify applicable funding agency here. If none, delete this.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m^2” or “webers per square meter”, not “webers/m^2”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm^3”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not

"Eq. (1)" or "equation (1)", except at the beginning of a sentence: "Equation (1) is . . ."

D. LaTeX-Specific Advice

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don't use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

BibTeX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LaTeX does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

E. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [7].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 1", even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a]Sample of a Table footnote.

Fig. 1. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use "Ref. [3]" or "reference [3]" except at the beginning of a sentence: "Reference [3] was the first ..."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors' names; do not use "et al.". Papers that have not been published, even if they have been submitted for publication, should be cited as "unpublished" [4]. Papers that have been accepted for publication should be cited as "in press" [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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